An Experimental Project on the Effectiveness of Oregano and Tanglad Spray as a Natural Mosquito Repellent

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CHAPTER I

INTRODUCTION

Background of The Project

Mosquitoes are considered one of the most dangerous insects in the world because they are known carriers of various diseases that affect millions of people each year (World Health Organization, 2023). In tropical countries like the Philippines, the warm climate and frequent rainfall create favorable conditions for mosquito breeding, making mosquito-borne diseases a serious public health concern (World Health Organization, 2023). Diseases such as dengue, malaria, and chikungunya can be transmitted through mosquito bites, which may lead to severe illness and even death if not properly prevented. Since mosquitoes are commonly found in homes, schools, and outdoor environments, people are constantly exposed to the risk of mosquito bites in their daily activities. Because of this, effective mosquito control is essential in maintaining a safe and healthy environment.

Many individuals rely on commercial mosquito repellents such as sprays, lotions, coils, and electric vaporizers to protect themselves. These products are typically made from synthetic chemical ingredients designed to repel or eliminate mosquitoes. While effective, frequent use of these chemical-based products may pose potential health

risks. Some users may experience skin irritation, allergies, headaches, or respiratory discomfort, especially when used in enclosed spaces (Environmental Protection Agency, 2022). Additionally, the continuous use of chemical repellents may contribute to environmental pollution, as harmful substances can accumulate in the air, water, and soil. The cost of these commercial products can also be a concern, particularly for families with limited financial resources. Because of these issues, there is a growing interest in developing natural, safer, and more affordable alternatives.

Plants are known to contain natural compounds and essential oils that can repel insects. Herbal plants have long been used in traditional practices as natural insect repellents due to their strong scents and bioactive components that interfere with the sensory systems of insects (National Institutes of Health, 2021). These natural substances can effectively repel mosquitoes without causing harmful effects to humans when properly used, making them suitable for sustainable and eco-friendly solutions.

Tanglad (Cymbopogon citratus), commonly known as lemongrass, is one of the plants recognized for its insect-repelling properties. It contains compounds such as citronellal, citronellol, and geraniol, which produce a strong citrus-like scent that mosquitoes find unpleasant (Food and Agriculture Organization, 2020). These compounds can disrupt the ability of mosquitoes to detect human presence, making

tanglad a widely used ingredient in natural insect repellents. It is also readily available, affordable, and easy to prepare, making it practical for everyday use.

Another plant with similar properties is oregano (*Origanum vulgare*), which contains active compounds such as carvacrol, thymol, and terpenes. Carvacrol and thymol are classified as monoterpenoid phenols, a group of naturally occurring organic compounds known for their strong antimicrobial and insect-repelling properties. These compounds are characterized by their aromatic structure and ability to disrupt biological processes in microorganisms and insects. Terpenes, on the other hand, are a large class of organic compounds produced by plants, commonly responsible for their distinct aroma and defense mechanisms against pests.

These bioactive compounds play an important role in the plant's natural protection system. Carvacrol and thymol have been shown to affect the nervous and sensory systems of insects by disrupting their olfactory receptors, which are responsible for detecting human scent. As a result, mosquitoes experience difficulty in locating human hosts, leading to avoidance behavior. In practical applications, these compounds are widely used in natural insect repellents, disinfectants, and medicinal products due to their effectiveness and relatively low toxicity. Because oregano is commonly grown and easily accessible, it serves as a practical and sustainable source of these natural repellent compounds (National Institutes of Health, 2021).

Combining oregano and tanglad may enhance the effectiveness of a mosquito repellent due to the presence of multiple active compounds that work together to repel insects. When formulated into a spray, the extract can be conveniently applied in areas where mosquitoes are commonly found, both indoors and outdoors. Compared to chemical-based repellents, a plant-based spray is potentially safer for regular use and more environmentally friendly.

Along with these considerations, this project aims to develop a natural mosquito repellent spray using oregano and tanglad extracts and experimentally evaluate its effectiveness in reducing mosquito presence (World Health Organization, 2023). Specifically, it seeks to determine whether the combined plant extracts can serve as a safe, affordable, and eco-friendly alternative to commercial mosquito repellents. Through this project, the researchers aim to contribute to the development of sustainable solutions for mosquito control while promoting health safety and environmental protection.

Statement of the Problem

Mosquitoes continue to be a common problem in many environments such as homes, communities, and outdoor areas. These insects not only cause discomfort but also carry harmful diseases such as dengue, malaria, and other infections that may affect human health. Because of this, many people use commercial mosquito repellents to protect themselves from mosquito bites. However, most of these repellents contain chemical ingredients that may cause skin irritation, allergies, breathing problems, and environmental pollution when used frequently. In addition, some commercial products are expensive and may not always be available for regular use.

Because of these concerns, there is a need to develop a natural, safe, and affordable mosquito repellent that can be used in everyday environments. Herbal plants such as oregano and tanglad are known to contain natural compounds that may help repel insects, but their effectiveness as a mosquito repellent spray needs to be tested.

This study aims to determine the effectiveness of oregano and tanglad spray as a mosquito repellent and to find out if it can be used as an alternative to chemical-based products.

Specifically, this study seeks to answer the following questions:

1. Does the oregano and tanglad spray reduce the presence of mosquitos in the selected area?
2. How effective is the oregano and tanglad spray in repelling mosquitos based on observed changes in mosquito activity
3. To what extent is the oregano and tanglad spray effective as a mosquito repellent?

THEORETICAL FRAMEWORK

Plant Secondary Metabolite Theory

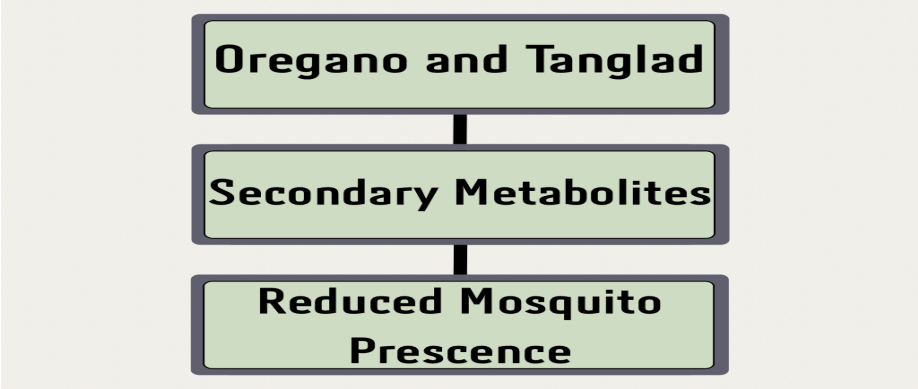


Figure 1. Diagram of Plant Secondary Metabolite Theory.

This project is anchored on the Plant Secondary Metabolite Theory, which posits that plants produce chemical compounds, called secondary metabolites, to protect themselves from predators, pests, and harmful microorganisms. These compounds are not directly involved in the plant's growth or reproduction, but serve as defense mechanisms that deter insects, fungi, and other threats. Examples of secondary metabolites include terpenes, phenols, alkaloids, and essential oils, many of which have demonstrated insect-repelling properties.

In the context of this project, oregano (Origanum vulgare) produces secondary metabolites such as carvacrol, thymol, and terpenes, while tanglad (Cymbopogon

citratus) contains citronellal, citronellol, and geraniol. These compounds act as natural repellents by disrupting the sensory mechanisms of mosquitoes, particularly their ability to detect human presence through scent. By utilizing these natural plant defenses, the oregano and tanglad spray is expected to reduce mosquito activity and presence in treated areas, providing a safe, environmentally friendly, and cost-effective alternative to chemical repellents.

Significance of the Project

This study will be beneficial to different groups of people because it aims to develop a natural mosquito repellent spray using oregano and tanglad as an alternative to chemical-based repellents. The results of this study may help provide a safer, affordable, and environmentally friendly solution for controlling mosquitoes in different environments.

Individuals and households. This study would give them an alternative to chemical mosquito repellants because the use of oregano and tanglad spray may help reduce mosquito presence at home and in surrounding areas. This may help prevent mosquito bites and reduce the risk of mosquito-borne diseases such as dengue and malaria while avoiding the possible harmful effects of chemical repellents.

Saint Mary's Angels College Of Valenzuela. This study would benefit the community of the school because it promotes the use of natural and locally available plants as an alternative to commercial mosquito repellents. Using plant-based products may help reduce environmental pollution and encourage safer practices in controlling insects.

Students. This study would help them increase their knowledge about the use of herbal plants in making useful products. This study may also help students understand the importance of using natural resources in a responsible and practical way.

Teachers and researchers. This study would help serve as a reference for future research related to natural insect repellents, herbal extracts, and eco-friendly products. The study may also help in developing improved formulas using other natural ingredients.

Future researchers. This study would benefit them with providing basic information about the effectiveness of oregano and tanglad as mosquito repellents. This may help them create better studies, experiments, or products related to natural pest control.

The environment. This study would help lessen the spread of chemicals in the environment. Plant-based repellents do not release harmful chemicals into the air, water, or soil; thus, using natural ingredients will help reduce pollution and support environmental protection.

Overall, this project may contribute to health safety, environmental awareness, and the development of natural alternatives that can be used in everyday life to control mosquitoes.

Scope and Limitations:

This study focuses on developing and evaluating a natural mosquito repellent spray using oregano and tanglad extracts. The research specifically investigates the spray's effectiveness in reducing mosquito presence in a selected area through direct

observation over a two-day period. The study emphasizes the preparation process, application, and observed repellent effects, aiming to provide a safe, affordable, and environmentally friendly alternative to chemical products. However, the experiment was limited to a single location and short observation period, which may not represent all environmental conditions. Only visual observations were conducted without quantitative mosquito counts or species identification, and the duration of effectiveness may vary depending on factors such as weather, wind, and extract concentration. Additionally, the inclusion of alcohol as a preservative may slightly affect the naturalness of the product, and no direct comparison with commercial repellents was made.

The project relies on qualitative observation, which may be subject to researcher interpretation. Additionally, the study does not include a control group or comparison with commercial mosquito repellents, which may limit the generalizability of the findings

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

FOREIGN REVIEW OF RELATED LITERATURE AND STUDIES

Supports the Use of Oregano in Mosquito Repellent

Khan et al. (2021) investigated the larvicidal, adulticidal, and repellent properties of oregano (*Origanum vulgare*) essential oil against *Aedes aegypti* mosquitoes. The study identified bioactive compounds such as carvacrol and thymol, which interfere with the sensory and physiological functions of mosquitoes, reducing their ability to detect and approach humans. The findings suggest that oregano essential oil is an effective, natural, and environmentally friendly alternative to synthetic chemical repellents. This supports the present study by providing a scientific basis for using oregano as a key ingredient in the mosquito repellent spray.

Validates the Safety and Natural Effectiveness of Oregano

Arumugam, Swamy, and Sinniah (2016) examined the bioactive compounds present in oregano, including carvacrol, thymol, and α-terpinene, and their role in repelling mosquitoes. The study emphasized that plant-based repellents are safer alternatives to chemical products, as they pose fewer risks to human health and the

environment. This supports the present study by confirming that oregano is both safe and effective as a natural mosquito repellent.

Supports the Use of Tanglad in Mosquito Repellent

Peach et al. (2019) evaluated the effectiveness of lemongrass (*Cymbopogon citratus*) essential oil in both laboratory and field experiments against *Aedes aegypti* mosquitoes. The results showed that lemongrass oil significantly reduced mosquito attraction by interfering with their host-seeking behavior. This supports the present study by confirming that tanglad is an effective natural mosquito repellent.

Supports the Combined Use of Oregano and Tanglad

Tolentino et al. (2023) assessed the repellency of different combinations of oregano and lemongrass extracts. The mixture containing 25% oregano and 75% lemongrass achieved a 100% repellency rate. The study concluded that the plant-based formulation was comparable to synthetic repellents. This supports the present study by providing evidence that combining oregano and tanglad enhances mosquito repellency.

Duration of Effectiveness of Plant-Based Mosquito Repellents

Studies have also examined the duration of effectiveness of plant-based mosquito repellents. According to Peach et al. (2021), lemongrass (*Cymbopogon citratus*) essential oil can effectively repel mosquitoes for approximately 2 to 4 hours, depending

on factors such as concentration and environmental conditions. The study explained that natural repellents tend to evaporate more quickly compared to chemical-based products, which may affect the length of their protective effect.

Similarly, Tolentino et al. (2023) reported that a combination of oregano (*Origanum vulgare*) and lemongrass extracts demonstrated strong mosquito repellency during the initial period of application, particularly within the first hour. However, the effectiveness gradually decreased over time. The study emphasized that reapplication may be necessary to maintain consistent protection against mosquitoes.

LOCAL REVIEW OF RELATED LITERATURE AND STUDIES

Confirms Oregano as a Safe and Effective Natural Repellent

Crisostomo et al. (2024) reported that oregano essential oil contains compounds that effectively repel mosquitoes and can serve as a safer alternative to chemical-based repellents. This supports the present study's use of oregano as one of the main ingredients in the mosquito repellent spray.

Provides Rationale for Combining Tanglad and Oregano

Stafford (2023), published in the *Ascendens Asia Journal*, explained that tanglad contains citronella and citral, which help mask human scents that attract mosquitoes, while oregano contains carvacrol and thymol that enhance insect-repelling effects. The

combination of these plants produces a synergistic effect that increases repellency, particularly against *Aedes aegypti*. This supports the formulation of the present study's mosquito repellent using both oregano and tanglad.

Provides Experimental Evidence for Effectiveness

De Jesus et al. (2023) conducted an experimental study using tanglad and oregano extracts and found that both plants exhibited mosquito-repellent properties. The study also showed that effectiveness increased as the concentration of the extracts increased. This supports the present study by demonstrating the practical application of oregano and tanglad in mosquito control.

Validates the Combination of Oregano and Tanglad

The study titled *Utilization of Indigenous Plants as Mosquito Repellent* (2019) investigated the use of oregano and tanglad extracts in repellent candles. The results showed that the combination of both extracts achieved 100% repellency within 15 minutes, outperforming individual extracts. This supports the present study by confirming that combining oregano and tanglad is more effective than using either plant alone.

Confirms Household Application and Effectiveness

Enad (2023) evaluated oregano and lemongrass extracts as organic household insect repellents. The findings showed that the combination of oregano and tanglad was

highly effective in reducing mosquito presence. This supports the present study's objective of developing a practical, safe, and effective mosquito repellent for everyday use.

CHAPTER III

METHODOLOGY

RESEARCH DESIGN

This study will use an experimental qualitative research design to determine the effectiveness of oregano and tanglad spray as a mosquito repellent. The study focuses on observing changes in mosquito presence and behavior before and after the application of the spray. A qualitative observational approach is used to describe mosquito activity based on visible patterns rather than numerical measurements, allowing the researchers to assess the practical effectiveness of the spray in a natural environment.

VARIABLES OF THE PROJECT

Independent Variable: Oregano and Tanglad Spray

Dependent Variable: Mosquito Presence and Activity

Controlled Variables: Time of observation, location, environmental conditions.

DATA RECORDING MATRIX

According to the World Health Organization (2009), field studies often utilize landing rate techniques, such as the Human Landing Catch (HLC) method, which measures the number of mosquitoes that land on a subject within a specified time period. This method is widely used in open environments because it focuses on mosquito interaction rather than total population, making it more practical and reliable under uncontrolled conditions.

Number of Mosquito Landings (within 5 minutes)	Interpretation
0 – 1	Highly Repellent
2 – 4	Repellent
5 – 8	Moderately Repellent
9 and above	Not Repellent

Figure 3.0 Measurement Table of Mosquito Presence Level.

MATERIALS AND EQUIPMENT

The following materials will be used in preparing the oregano and tanglad mosquito repellent spray:

- 40.5 grams Fresh oregano leaves
- 12 grams Fresh tanglad (lemongrass) leaves
- 4 cups Clean water
- Alcohol (optional, for preservation)
- Spray bottle
- Kitchen Knife
- Strainer
- Measuring cups

These materials will be chosen because they are safe, affordable, and easy to find.

Phase I: Preparation of Oregano and Tanglad Leaves

1. Fresh oregano leaves were collected.
2. Fresh tanglad leaves were collected.
3. The collected leaves were washed thoroughly using clean water.
4. The leaves were prepared and set aside for the extraction process.

Phase II: Process of Making the Oregano and Tanglad Spray

1. Clean water was boiled in a container.
2. Once the water reached boiling point, the prepared oregano and tanglad leaves were added.
3. The mixture was allowed to boil for a few minutes to extract the active compounds.
4. After boiling, the mixture was removed from heat and allowed to cool.
5. The liquid extract was transferred into a clean container
6. A small amount of alcohol was added as a preservative.
7. The solution was poured into a spray bottle.
8. The spray bottle was properly labeled.
9. The oregano and tanglad spray was ready for use in the experiment.

SCHEMATIC DIAGRAM

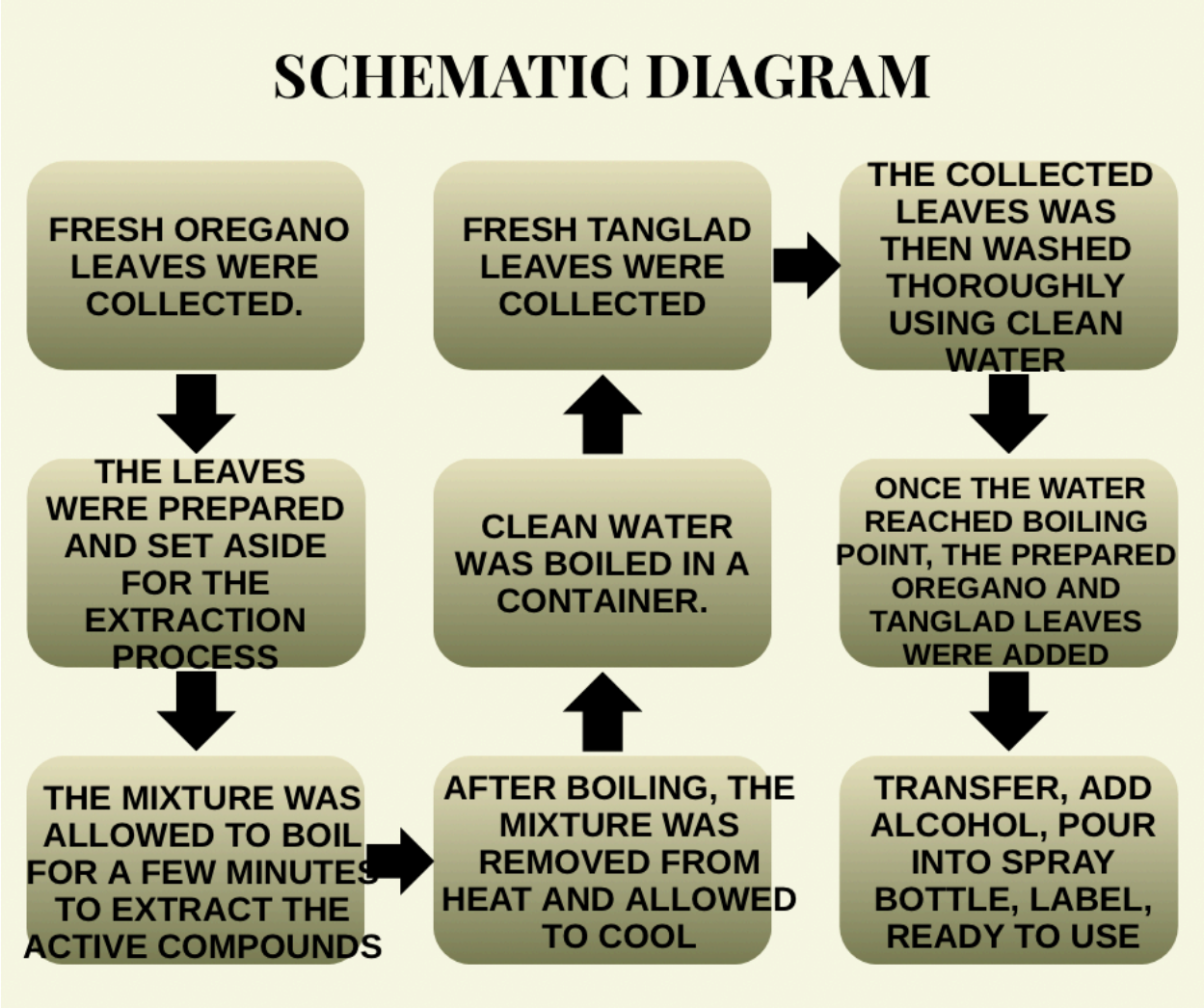


Figure 3.1Schematic Diagram of Process in Making Repellant Solution.

Data Gathering Procedure

The researchers will gather data through direct observation of mosquito presence in the selected area over a period of two days. Observations will be conducted at the same time each day to ensure consistency of environmental conditions.

On the first day, the researchers will observe and record the presence of mosquitoes in the area prior to the application of the oregano and tanglad spray. After the initial observation, the spray will be applied, and the area will be observed again to determine any changes in mosquito presence.

On the second day, the same procedure will be repeated to validate the consistency of the results. All observations will be documented using a structured observation sheet, focusing on the presence and activity of mosquitoes within the area.

To ensure consistency and reliability in the qualitative data, mosquito presence will be categorized based on observed activity. High mosquito activity is described as frequent flying, hovering, and landing of mosquitoes within the area. Moderate mosquito activity refers to occasional mosquito presence with fewer visible movements. Low mosquito activity indicates minimal to no visible mosquitoes observed in the surroundings.

The collected qualitative data will be analyzed by comparing the observed conditions before and after the application of the spray. This will allow the researchers to determine whether the oregano and tanglad spray is effective in reducing mosquito activity and presence in the selected area.

Ethical Considerations

The researchers will make sure that the study will be safe for people and the environment. Only natural ingredients such as oregano and tanglad will be used to avoid harmful effects. Permission will be asked before conducting the experiment in the neighborhood. The researchers will make sure that the activity will not disturb other people or damage property. All data gathered during the study will be used only for research purposes.

CHAPTER IV

RESULTS AND DISCUSSION

Observation

	TIME	PRESENCE OF MOSQUITOES
NIGHT 1	6 PM	Not Repellent
	10 PM	Highly Repellent
NIGHT 2	6 PM	Not Repellent
	10 PM	Highly Repellent

During the preparation of the oregano and tanglad spray, the researchers observed that the extraction process produced a clear liquid with a strong and distinct natural aroma. The procedure was simple and manageable, as the materials used were readily available and the steps were easy to follow. The mixture was successfully filtered, resulting in a liquid extract suitable for spray application. The addition of alcohol helped enhance the extraction of active plant compounds and served as a preservative for the

solution. The prepared spray was then transferred into a labeled spray bottle for experimental use.

During the experimental phase, observations were recorded using the Data Recording Matrix under two conditions: no repellent and with oregano and tanglad spray. Under the no repellent condition, mosquito activity was observed to be high, characterized by frequent flying, hovering, and landing of mosquitoes in the selected area.

After the application of the spray (with repellent condition), mosquito activity showed a gradual decrease based on the matrix categories. The level of activity changed from high to moderate, and eventually to low, indicating a significant reduction in mosquito presence within the treated area.

Mosquitoes were observed to exhibit avoidance behavior, as fewer insects remained or approached the sprayed surroundings compared to the no repellent condition. The strong natural aroma of the spray appeared to disperse effectively in the environment, contributing to its repellent effect.

Based on the matrix recordings, the treated area maintained low mosquito activity for approximately four hours after application. After this period, mosquito activity

gradually increased, approaching the levels observed under the no repellent condition as the scent diminished.

Results and Discussion

The researchers found that the preparation of the spray was simple and utilized readily available materials. The process produced a clear liquid that functioned effectively when applied using a spray bottle. The addition of alcohol enhanced the extraction of active plant compounds and helped preserve the solution. The final product had a strong natural aroma derived from oregano and tanglad.

Prior to the application of the spray, high mosquito activity was observed in the selected area, characterized by frequent flying and hovering of mosquitoes. After applying the oregano and tanglad spray, mosquito activity decreased to moderate and eventually low levels, indicating a noticeable reduction in their presence.

The reduction in mosquito activity can be attributed to the presence of secondary metabolites such as citronellal, citronellol, carvacrol, and thymol. These compounds interfere with the sensory mechanisms of mosquitoes, particularly their ability to detect human presence through scent.

The observed repellent effect lasted for approximately four hours, after which mosquito activity gradually returned as the scent weakened. The consistent decrease in mosquito activity across both days suggests that the oregano and tanglad spray is effective as a natural mosquito repellent.

Overall, the findings demonstrate that the combination of oregano and tanglad can significantly reduce mosquito activity, making it a practical, affordable, and environmentally friendly alternative to chemical-based repellents.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

Conclusion

The researchers found that the preparation of the oregano and tanglad spray was simple and practical, utilizing readily available and affordable materials. The extraction process produced a clear and uniform liquid that functioned effectively when applied using a spray bottle. The inclusion of alcohol contributed to better extraction of active plant compounds and served as a preservative, helping maintain the stability of the solution over time. The final product emitted a strong natural aroma derived from oregano and tanglad, indicating the presence of bioactive compounds responsible for its repellent properties.

Based on the Data Recording Matrix, a clear comparison was established between the no repellent and with repellent conditions. Prior to the application of the spray (no repellent), mosquito activity in the selected area was consistently categorized as high, as evidenced by frequent flying, hovering, and landing of mosquitoes. This confirms that the area naturally had a high level of mosquito presence in the absence of any control measure.

After the application of the oregano and tanglad spray (with repellent), a significant change in mosquito activity was observed. The recorded data showed a gradual decrease from high to moderate, and eventually to low levels of activity. This progressive decline indicates that the spray had a direct and observable effect in reducing mosquito presence. Compared to the no repellent condition, the treated area showed fewer mosquitoes and reduced movement, supporting the effectiveness of the spray as reflected in the matrix.

The observed reduction in mosquito activity can be attributed to the presence of plant secondary metabolites such as citronellal, citronellol, carvacrol, and thymol. These compounds are known to interfere with the sensory mechanisms of mosquitoes, particularly their ability to detect human hosts through scent. As a result, mosquitoes were less likely to approach or remain in the treated area, which explains the transition from high to low activity levels recorded in the matrix.

In addition, the data revealed that the repellent effect of the spray lasted for approximately four hours, during which mosquito activity remained at a low level under the with repellent condition. After this period, mosquito activity gradually increased, shifting toward moderate to high levels, similar to those observed in the no repellent condition. This suggests that the effectiveness of the spray is time-dependent and decreases as the active compounds evaporate or dissipate in the environment.

The consistency of the results across the two-day observation period further strengthens the reliability of the findings. The repeated pattern of high activity in the no repellent condition and reduced activity (moderate to low) in the with repellent condition demonstrates that the oregano and tanglad spray has a consistent and measurable impact on mosquito behavior.

Overall, the findings of the study, as supported by the Data Recording Matrix, demonstrate that the combination of oregano and tanglad is effective in significantly reducing mosquito activity. The spray can therefore be considered a practical, affordable, and environmentally friendly alternative to chemical-based mosquito repellents, especially for short-term use within a limited time frame.

Recommendation

For future research, it is recommended to conduct longer-term studies across various environments, both indoors and outdoors, to further validate the consistency and effectiveness of the oregano and tanglad spray. Researchers may also incorporate quantitative methods, such as mosquito traps, counting techniques, or landing rate measurements, to obtain more precise and objective data. In addition, testing the spray against different mosquito species may provide a broader understanding of its effectiveness in various conditions. Future researchers are also encouraged to explore alternative extraction techniques, such as steam distillation, solvent extraction, or other advanced methods, in order to obtain more concentrated and accurate extraction of active plant compounds. These improved techniques may enhance the overall quality, consistency, and effectiveness of the repellent.

Furthermore, exploring the use of fully natural preservatives could improve the eco-friendliness of the product, while experimenting with additional herbal extracts may enhance both the duration and strength of the repellent effect. To make the spray more user-friendly, adding a small amount of natural oil may help reduce the strong herbal scent while maintaining its effectiveness. For practical application, it is recommended to reapply the spray every three to four hours for continuous protection, and to encourage households and schools to prepare small batches for regular use. Promoting the use of

locally available plants as natural repellents can help reduce dependence on chemical-based products while supporting environmental sustainability and community-based solutions.

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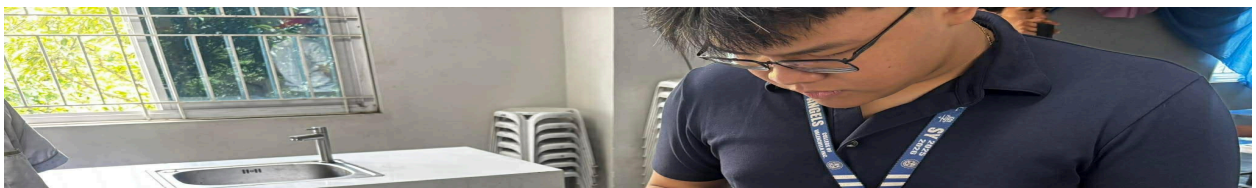
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APPENDIX A








ANGELS




DO NOT
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MODE
TARE


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PCS
ON/OFF
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Ming Heng Electronic Digital Scale















APPENDIX B



